

How would you visualise your data?

COMP5048/4448 Assignment 2

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I. DATA EXPLORATION AND PRE-PROCESSING

The dataset is collected from Ben Hamner on Kaggle [1], showing data about the US Presidential Election 2016. This event was a historic occasion that shocked many observers. Donald Trump prevailed in the election, contrary to popular expectations that Hillary Clinton would win handily. Both candidates used social media to interact with voters and sway public opinion, especially Twitter. This dataset offers a thorough compilation of tweets from both candidates during the election season.

We utilized some NLP techniques to clean the text such as removing URLs, stopwords (such as is, and, are, etc.), meaningless words, punctuation, etc. This step was to avoid the value for these words override the visibility of the other significant words [2].

II. TASK A – THEMATIC ANALYSIS AND ENGAGEMENT

A. Visualization Process Overview

In this part, we first find ways to identify different themes of discussion, then visually represent different themes of discussion. After that, we display the level of engagement (retweets and likes) for each theme within the same visualization.

B. First Visualization: Explore Words and Engagement

Before doing this visualization, we integrated all text, clean it as mentioned above and find the sum retweet_count and favorite_count for each word to find the engagement when that word appears in the sentence.

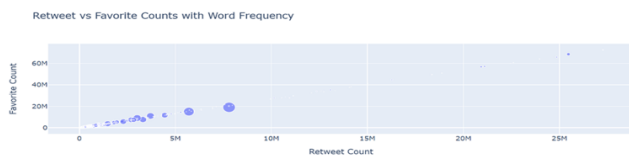


Fig. 1. Retweet vs Favorite Counts with Word Frequency Visualization

This visualization uses scatter plot. The choice of scatter plot helps showing the spreading of each datapoint, in this case, word, more clearly. It also helps find patterns in data. The axes including favorite count and retweet count are used because this visualization chart focuses on the popularity of each word; these two attributes need to be clearly displayed. This can firstly help users clearly see the distribution of words in "favorite count" and "retweet count" and secondly help users more easily identify words with more "favorite count"

and "retweet count" so as to select meaningful and popular words. In this way, keywords can be found.

A certain visual variable called size is used. The size of a data point represents the number of times the word was mentioned in all tweets. The size of the data point clearly distinguishes the frequency of word occurrence, making it easier for users to understand the frequency of occurrence of specific words. Certain interactions include zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because zooming in and out of images and moving images help users understand the overall distribution of the data and the details they want to focus on. When the mouse is hovering over a specific data point, the data point's favorite count, retweet count, count and content of the word is shown, helping users to identify the words they want.



Fig. 2. Retweet vs Favorite Counts with Word Frequency Visualization with Hover Interaction

We can identify a series of words with high occurrence rates and classify them into the same theme, for example, the following figures show 4 found keywords: 'women' in 'Women Right', 'gun' in 'Crime', 'economy' in 'Economist', 'shooting' in 'Crime'.

In the same way, we scanned through the visualization, and got a whole set of topics with keywords, for example, 'Election' includes 'president', 'vote', 'poll', 'campaign', 'Business' includes 'business', 'company', 'full-time', 'wage', 'jobs', 'wealth', 'foundation', 'charitable', 'charity', 'work', 'job', 'pays', 'manufacturing', 'outsourcing', 'Income' includes 'income', 'incomes', 'Crime' includes 'crime',

'criminal', 'shooting', 'shoot', 'attack', 'violence', 'violent', 'murders', 'prisoners', etc.

C. Second Visualization: Explore Themes and Engagement By Word

After getting the list of keywords and identified themes from the 1st visualization, we decided to plot the words again but with different themes

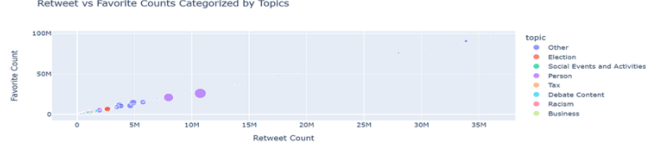


Fig. 3. Retweet vs Favorite Counts with Themes Visualization

This visualization uses scatter plot. The choice of scatter plot helps showing the spreading of each datapoint, in this case, word, more clearly. The axes including favorite count and retweet count are used because this visualization also shows the popularity of each word, like the last one. This can help users to see distribution of data points on these two attributes clearly, seeing each word's favorite count and retweet count. Certain visual variables including color of datapoints, size of data points were used, the color of datapoints shows the themes of each word, helping users understand which theme is more popular and is more mentioned. They would help users to identify which theme's words are mentioned more with "count" attribute, and which of them have more favorite count and retweet count.

Certain interactions include filtering by theme, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because zooming in and out of images and moving images help users understand the overall distribution of the data and the details they want to focus on. When the mouse is hovering over a specific data point, the data point's topic, retweet count, favorite count, count and word are shown.

D. Third Visualization: Different Themes with Engagement Level

This is the additional visualization that summarize the existing themes with their engagement. First, we process the data by calculating the average values: all favorite counts of each tweet in each theme are summed up and calculate the average to get the "average favorite count", the retweet count is calculated into "average retweet count" of each theme by the same method. The generated visualization shows the average value of favorite count and retweet count for each theme, with attributes including "theme", "average favorite count", "average retweet count".

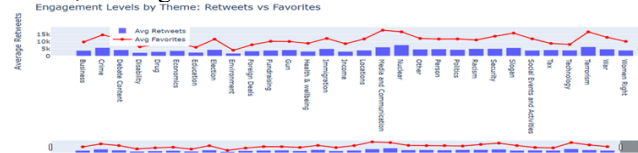


Fig. 4. Engagement Levels by Theme: Retweets vs Favorites Visualization

This visualization uses line charts and bar charts. The line charts represent average favorites for each theme, and bar charts represent average retweets for each theme. The line chart and the bar chart clearly differentiate the retweets and

favorites. First, they make it easier for users to compare their changes within one theme. Secondly, bar charts and line charts help see the difference and changes between themes. Axes including values and themes are used because this visualization depicts retweets and favorites for each theme directly helps users see the average retweet and favorites of each theme more clearly. By showing the "average favorite count" and "average retweet count" of each theme, we can show the level of engagement of each theme.

Certain interactions include sliders, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because for slides, we can zoom in on the theme more conveniently. For mouse hovering, when a mouse is hovering over a specific theme, the theme's name, average retweets and average favorites are shown. The zooming in and out of images and moving images are of the same effect as that of the previous visualization.

E. Conclusion

In conclusion, for the visualization in Fig.1 and Fig.3 in part A, we used visualization tools to help discover and identify different discussion topics. For using evidence to define groups, as nominal data in visualization is complicated, we found some evidence to help us define the groups, that is, to find keywords and group similar topics together [3]. This evidence is presented through our visualizations Fig.1 and Fig.3, including patterns, trends, and themes in the data. Fig.1 shows the trend of the data in favorite count and retweet count, which basically shows a linear increase, which means that the higher the level of engagement of the data points in the upper right corner, the more priority should be given to the words. In Fig.3, after the themes are separated, the data points are colored as the classification of the themes, so that it is easier to see which themes have a higher level of engagement. Through these two figures, we completed the discovery and explanation of the grouping.

The visualization in Fig.4 visually represents different themes of discussion in a visualization. In this visualization, for each theme, level of engagement, which are represented with average retweets and likes are shown. It can be seen tha Crime, Media and communication and Terrorism are the themes with higher levels of engagement. For example, as is shown in the figure, Crime theme has 5742754 average retweets and 14.52k average favorites. Disability, Education and Environment are the themes with lower levels of engagement. For visualization design and interpretation, our visualization matches the task requirements. Our visualization clearly shows the distribution and engagement of the topic. The reason and intention behind this design is that the line chart and bar chart represent the average retweets and average favorites count respectively, so that we can see the difference between them in different topics and compare their relationship in the same topic (it can be observed that the average retweets and average favorites count basically increase and decrease at the same time) [4].

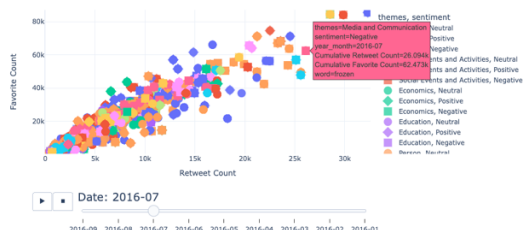
III. TASK B – SENTIMENT ANALYSIS

A. Visualization Process Overview

Through the word visualization in task A, we also obtained some emotional words and phrases. Then, we generated

B. First Visualization: Explore Themes and Sentiment keywords

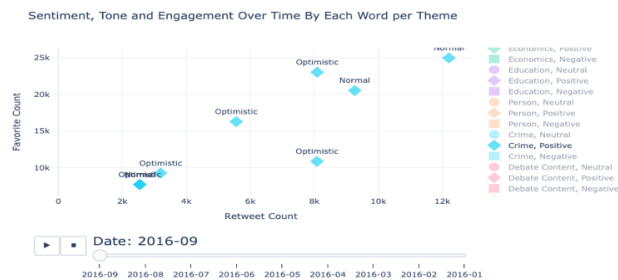
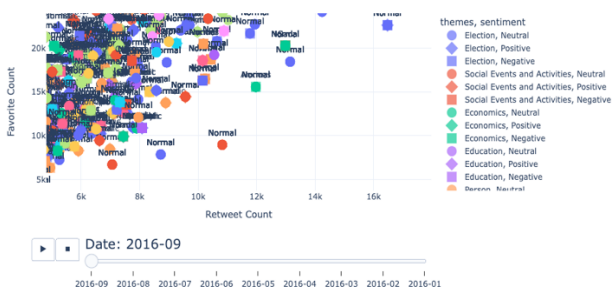
Sentiment and Engagement Over Time By Each Word per Theme



The visualization uses a plot chart. The axes include favorite count and retweet count. It directly shows the audience's engagement with the words.

C. Second Visualization: Explore Themes, Sentiment, and Tone keywords

Sentiment, Tone and Engagement Over Time By Each Word per Theme



Scatter plot is used because it is easier to see the distribution of data points, in this case, the popularity of each keyword. The axes include favorite count and retweet count. These are 2 values taken from the tweet contain the word, so that 2 words can be at the same position but different theme, sentiment, and tone. Also, therefore, the word can be duplicated because it appeared in different time and different theme. By using this, we can visualize how each word engagement difference when they are in different context.

Certain interactions include time slides, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because for the time slide, it helps to show how tweet changes throughout the campaign. The mouse hovering over a specific data point shows the data point's theme, sentiment, retweet count, favorite count, tone, time and text. The zooming in and out of images and moving images are of the same effect as that of the previous visualization.

After getting the tone and sentiment keywords, we visualize how each tweet sentiment and tone of 2 candidates evolve throughout the campaign



Fig. 7. Engagement Metrics by Date per Candidate Visualization (with Hover Interaction)

Scatter plot is used to show how data points of a tweet are spread on the theme and audience engagement with respect to different times. The axes let users directly see audience engagement and different themes. In this way the change of audience engagement of a certain theme throughout the campaign can be shown clearly.

Certain visual variables including color of data point and shape of datapoint were used, the color of data point indicates tweets from Hillary or Trump, to help users filter the different candidate's tweets to better understand and compare audience engagement of their tweet's. The shape indicates sentiment of "positive", "negative" or "neutral", letting user identify the different sentiment or filter the wanted tweets with certain sentiment to better analyze the data.

Certain interactions include a time slide, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because for the time slide, it helps show how the tweets change throughout the campaign. For the mouse hovering over data points, the data point's candidate, sentiment, theme, audience engagement, tone, retweet count, favorite count, time and text are shown, allowing users to have a more detailed understanding of each data point. The zooming in and out of images and moving images are of the same effect as that of the previous visualization.



Fig. 8. Tone Distribution over Time by Candidate with Audience Engagement Visualization (with Hover Interaction)

Another similar visualization is produced, but this is to view the change through each day so that the date is not shortened to month only. The reason why using scatter plot is the same as the previous one. However, the axes are different. They include audience engagement (favorite count and retweet count) and tone with candidates. It directly shows the audience's engagement with the different candidates and their tones.

E. Fourth and Fifth Visualization: Change In each Theme Sentiment and Tone

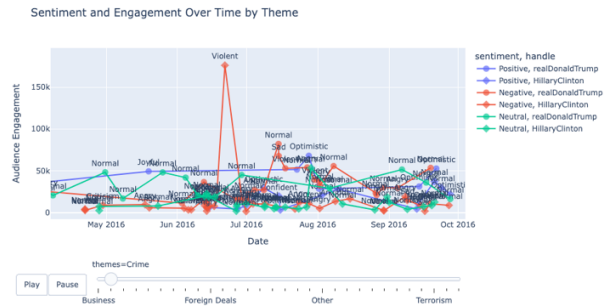


Fig. 9. Sentiment and Engagement Over Time By Theme Visualization

The line chart above visualizes how audience engagement on different topics changes over time. Existing techniques use line graphs to show the distribution of emotions, and these techniques are well suited to the time trends discussed above, proving their effectiveness [5]. The topic selection is implemented by a slide, and each data point in the line chart is the audience engagement of a tweet. The line chart has three colors, representing three sentiments, and the data point has two shapes, representing two candidates.

The visualization uses a line chart. The axes include audience engagement (favorite count and retweet count) and time by month. It directly shows for each theme, how the audience's engagement changes with the different candidates and their sentiments.

Certain visual variables including color of the data point (with the line) and shape of the data point. The color represents the sentiment of each datapoint (a tweet). The shape represents the candidate. The color and the shape of the data points can show each tweet's sentiment and candidate clearly, helping users to spot the sentiment change and more popular themes at different time periods.

Certain interactions include a slide for theme, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because for the slide for theme, it shows the tweet's changes throughout the campaign for each theme separately, making it easier to see details for one theme. When the mouse is hovering over a specific data point, the data point's sentiment, candidate, theme, time, audience engagement, tone, retweet count, favorite count and text are shown. The zooming in and out of images and moving images are of the same effect as that of the previous visualization.

Sentiment and Theme Analysis with Engagement Levels for Both Candidates

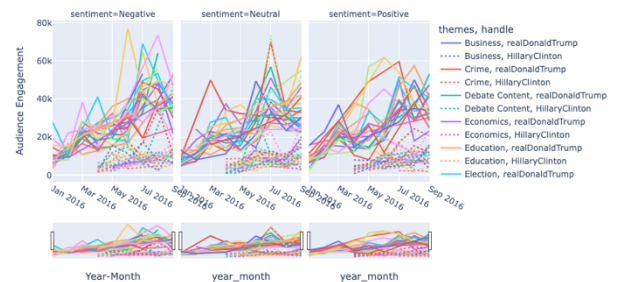


Fig. 10. Sentiment and Theme Analysis with Engagement Levels for Both Candidates Visualization

This graph illustrates how the topics and sentiment patterns for both candidates changed throughout the campaign, as well as how audience engagement and sentiment were distributed for each theme. The graph's average audience engagement for each month illustrates how appealing the various themes are, the line style sets the contenders apart, and the colors denote the subjects of conversation. This graphic aids in comprehending how the audience's reaction and each candidate's sentiment changed over time. Sentiment is depicted in this visualization in the following ways: Different colors are used to distinguish between the three sentiment categories (positive, neutral, and negative). One color is used for "Positive," another for "Neutral," and a third for "Negative." We can easily recognize lines for various mood groups thanks to this color labeling. Candidates are distinguished by their line styles: It is simpler to examine trends in audience engagement among candidates in the same mood since candidate line styles (such as dashed and solid) distinguish each candidate's performance in various mood categories. Audience Engagement: Lines are a great way to show how audiences engagement levels for mood categories change over time. The lines' placement indicates the degree of audience participation for various moods, making it easier to determine which moods elicit greater levels of audience participation. This average is determined using data aggregation's grouped averaging method.

F. Conclusion

In conclusion, in task B, we used visualization to analyze trends of sentiment changes and topic changes with timeslide. Through features such as dynamic rotation and range selection, users can analyze data more flexibly and discover potential relationships and important information [6]. The visualization in Fig. 5 shows the level of engagement of words with different topics and different sentiments at different times. The visualization in Fig. 6 shows the level of engagement of words with different topics and different sentiments at different times. The visualization in Fig. 7 shows the level of engagement of tweets with different sentiments, tones, and topics at different times (monthly). The visualization in Fig. 8 shows the level of engagement of tweets with different sentiments, tones, and topics at different times (daily), creating visualizations for multivariable data with the scope of time [7].

For design and explain visualizations, to show the changing trends of candidate sentiment and topics. We chose Fig. 7 and Fig. 8 because they show the trend changes of sentiment and topics respectively. Fig. 7 shows all topics and gives the viewer an overall understanding of the month, while Fig. 8 only shows the topic of the day and focuses on showing the details of each day.

For visual evidence of sentiment changes, our visualization clearly shows the changes in candidate sentiment, including the overall trend of sentiment, such as changes in positive, negative or neutral sentiment. Visualizations intuitively show how sentiment changes over time or events. Fig. 7 and Fig. 8 show that most of the time is positive sentiment. In visualization in Fig. 7, Fig. 8, we can see that Trump is opposed to Hillary and her social events and has obtained a high level of engagement. For example, Trump's tweet on Fig. 8 obtained a level of engagement of 119k. In addition,

Trump often has a negative attitude towards the media. Many negative tweets were posted on June 19

For visual evidence of topic changes, similarly, our visualization clearly shows how topics change. For example, the amount of discussion or popularity of different topics changes over time. Fig. 6 Sentiment and Engagement Over Time by Each Word per Theme shows that the level of engagement gradually increases over time, reaching its highest level in May, June, and July, and then slowly declines.

For comprehensive display of sentiment and theme, Fig. 7 and Fig. 8 show the changes in sentiment and theme at the same time in the same visualization chart. This can help observe the relationship between the two, such as how a certain topic changes in sentiment, which can be clearly understood through Fig. 7. Or whether the change in sentiment is related to the change in theme, Fig. 8 can show the daily changes, as described above for the topic and sentiment of Trump's tweets.

IV. TASK C – CRITICAL MOMENT ANALYSIS

A. Hypothesis



Fig. 11. Screenshots of Critical Moment from Task B Visualizations

By analyzing the visualization in Task B, we noticed that Hillary's tweets about crime topics saw a significant increase in audience engagement between June and July 2016 for tweets containing positive sentiment keywords. Specifically, we looked at the data points for different sentiments (positive, neutral, negative) in the line chart and found that the line for positive sentiment was trending upward over that time period. At the same time, we deeply analyzed these high-engagement tweets and found that they contained positive keywords such as 'protection', 'peace', and 'sacred'. This suggests that tweets with positive sentiments received more attention during this period compared to the previous trend of tweets with negative sentiments receiving more engagement. Before that, the negative tone tweets have more engagement than the positive and neutral ones. The first positive tweet that marked the

increased engagement is 'I mourn for the officers shot while doing their sacred duty to protect peaceful protesters, for their families & all who serve with them.' Although this tweet expresses the mourning, it praises the contribution of the officers with more positive keywords such as protect, peaceful, sacred so that it tends to have more positive sentiment. Therefore, we come up with a hypothetic question:

What if Hillary gave a tweet with the same content but with more negative words (showing more negative sentiment)?

B. Experiment Method

In order to test for the hypothesis, we have come up with a new alternative tweet with the same content but with more negative keywords. The new tweet is:

'I mourn for the officers sad lost, brutal shot while trying to protect misguided protesters; this violent attack reveals society profound failure, leaving shattered families in deep despair.'

The idea for this alternative tweet was from the below visualization. We analyzed Hillary's previous negative tweets on crime topics, especially those with high audience engagement, and extracted negative keywords that appeared frequently and were associated with high engagement. For example, we found that words such as "tragic", "violent", "failure", and "desperate" appeared frequently in these tweets. Based on this, we added as many of these negative keywords as possible to the newly conceived alternative tweets, aiming to simulate a tweet with more negative sentiment to test our hypothesis. We would like to include as many of these words as possible in order to increase the predicted audience engagement of new tweet.

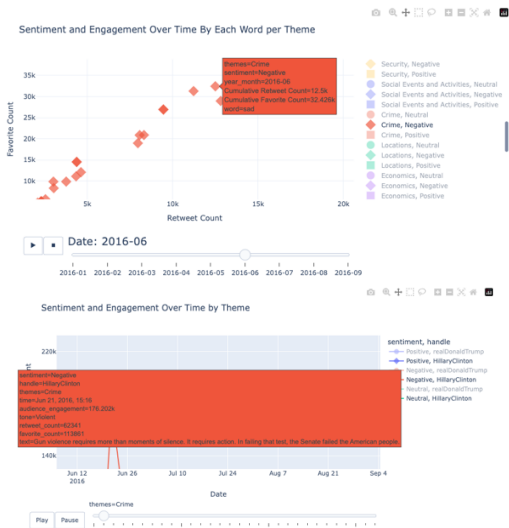


Fig. 12. Negative Keywords of Crime Theme (screenshots from task B visualization)

Specifically, we first preprocess the tweet text to remove stop words, punctuation marks, and meaningless words, retaining only keywords with actual meaning. Then, we calculate the average audience engagement corresponding to each word in all tweets. This is obtained by counting the engagement of tweets containing the word and averaging it. Next, for each

tweet, we calculate the average audience engagement for all keywords it contains as the predicted engagement for that tweet. We choose to use the average method to avoid bias caused by different tweet lengths and prevent tweets from being predicted to have higher engagement simply because they contain more words [8]. In this way, we ensure that the predicted engagement between different tweets is comparable.

C. Experiment Result

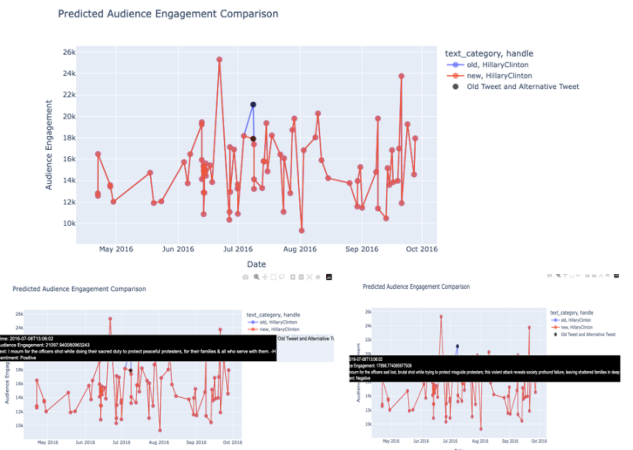


Fig. 13. Predicted Audience Engagement Change Comparison Visualization

This visualization is used to see the change in predicted audience engagement of tweets from Hillary Clinton and compare the change between the old and new tweet. Therefore, to visualize the change, we included the time of each tweet, the predicted audience engagement (because it's the main metric to compare), the text_category (to distinguish between new tweet and old tweet). Since it's the temporal analytics, using a line chart with x-axis is the time and y-axis is the predicted audience engagement is the most appropriate method. Because the text_category is the binary categorical data so we used the different hue color visual variable to distinguish between 2 categories. In order to spot 2 compared tweets, highlight interaction is used to highlight these 2 for users to easily identify which points are being compared. The interactive visualization is used in order to go deeper in details for analytical purposes. We used spanning, zooming, and hovering in this visualization because there is overlap between tweets with similar positions so that using zoom could help see those points separately. Moreover, hovering helps read the actual values of each point and more details about it to make a connection between the other details with the attributes visualized by visual variables.

Treemap of Words by Audience Engagement



Fig. 14. Treemap of Words by Predicted Audience Engagement Visualization

This visualization is used to visualize the average engagement of each word in 2 compared tweets (old content and new content). There is the hierarchical relationship between tweet category and the words in the tweets because the words form the sentences. Therefore, using treemap is possible here. The size visual variable is used to represent the predicted average audience engagement of each word because this is the ratio data that is better to be compared by size of each word block. The color visual variable is used to represent the categories of the tweets (old or new tweet). The hover user interaction is used to show more details about each word such as average engagement count, word, text category, etc.

D. Conclusion

As shown in the line chart, the blue line is the change with the old tweet. The previous one and the one after the target tweet are the negative ones, which have lower engagement than the target one. The red line is the new tweet. Since the text other than target tweet are kept so that 2 lines are overlapped except for the target tweet. We can see the new tweet has lower predicted audience engagement than the old one. In the tree map, the predicted total audience engagement of the new tweet is more than the old tweet; however, when moving through each word with hovering interaction, we've seen that most of the words have lower average audience engagement than the old tweet. Therefore, we supposed that since the new tweet has longer length than the old tweet so that the total value is bigger. After considering 2 visualizations, we made a conclusion that if Hillary posted with a more negative tone in that critical moment, the engagement will probably not be as high as the old positive tone.

V. EVALUATION

A. Set Up

Fig.1 is used to do the evaluation with other people to see if any improvement can be made. Since this is interactive visualization, it's more appropriate to conduct an interview with users to gather information. This method helps to discover facts and explore more explanations from users. In order to not cause users to have irrelevant evaluation about the plot, we decided to use a structured interview which contains a predefined set of questions [9].

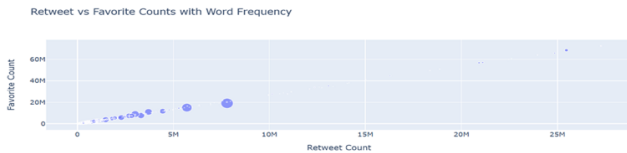


Fig. 15. Chosen visualization for Evaluation

The question list are as below:

Close-ended Questions

Question 1: Were you able to draw the main conclusions we wanted to convey through this visualization?

Question 2: Can you identify which keywords are most important in this visualization?

Question 3: Can you tell from the visualization which word has more audience engagement on social media?

Question 4: Can you understand what each data point size represents? For example, do they effectively indicate word frequency or engagement?

Open-ended Questions

Question 5: What difficulties did you encounter in understanding this visualization? What are some suggestions for improvement?

Question 6: On the whole, are these visualizations clear and intuitive, and are they effective in helping you understand the data? Please explain what you think.

This set of question contains both close-ended questions and open-ended questions. For close-ended questions, we expect the user to answer either Yes or No. Because this is an interview, we may ask user to clarify more about their choice. For open-ended questions, we allowed user to give any answer they can think of.

B. Interview Conduct

The participant with the interview was a student who is currently taking COMP5048 – Visual Analytics. With this participant, they have the basic knowledge about interactive visualization and data visualization. Therefore, it was assumed that there was no need to instruct the user basic step to do the interview and how to use the interactive visualization.

Below is their recorded response:

TABLE I. PARTICIPANT EVALUATION ANSWER

Question Number	Participant Answer
1	Yes, the visualization uses different visual variables to show audience engagement and keyword. I can see the popularity of keyword.
2	Yes, the visualization shows the most important keywords by their high frequency and the degree of interaction (likes, reposts). However, I cannot see each keyword clearly because I cannot zoom in and out or click.
3	Yes, through visual variables such as scatter plots and colors in the figure, I can intuitively see which keywords have received more attention and higher interaction volume.
4	I can understand. The size of the data point represents the word frequency. The size of the data point in the scatter plot represents the frequency of occurrence of the keyword in the tweet. The larger the size of the data point, the more times the keyword appears.
5	I think some data points overlap too much, making it difficult to distinguish different data points. More interactive features can be introduced, such as hovering the mouse to display more detailed information. Also, some data points are too small that I can't identify with bare eyes.
6	Overall, I find this visualization relatively clear, but some aspects, such as excessively overlapping data points, are not very clear and prominent, which affects the intuitiveness.

- Overlapped and small points: There are some small points that hard to identify with bare eyes. This may because the difference with size of each point (the word frequency)
- Unclear interaction: There is zooming interaction; however, it may cause the user a little hard to identify where to do the zooming. This might need more instruction from the group member.
- Good information conveyed: With chosen values on 2 axes and visual variables, the user can understand the main idea we want to convey through the visualization.

In order to improve the readability of the data points, we decided to add the word as text visual variable on top of each data points. Also, as mentioned by the participation, there is an overlap on each data point, we decided to move the text to the right with smaller font size to reduce the overlapping. For the unclear interaction, we interviewed again with participant and gave a clearer instruction on how they could do with the visualization interaction.

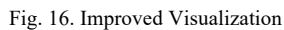


TABLE II. PARTICIPANT EVALUATION ANSWER ON IMPROVED VISUALIZATION

Question Number	Participant Answer
	received more likes and reposts than other words. I can easily tell which words have higher audience engagement by observing the position of the data points.
4	I can understand. The size of the data point represents the word frequency. The size of the data point in the scatter plot represents the frequency of occurrence of the keyword in the tweet. The larger the size of the data point, the more times the keyword appears.
5	Now I can clearly see the data points through interaction, so I think there may not be too many problems.
6	Overall, this visualization clearly shows the relationship between likes and retweets, and uses the position of the two axes to represent the popularity of different keywords. Now in areas with more data points, through the interactive sharing function, I can clearly zoom in and hover the overlapping areas to observe the data.

We want to sincerely thank Masahiro Takatsuka, Behroz Newaz Khan, and teaching team for all of their help and advice during this project. Additionally, we would like to thank our friend (Anonymous) for their assistance with the evaluation process. Lastly, we would want to express our gratitude to everyone who helped to successfully complete this report.

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